MARGARET MURAKAMI

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San Jose, CA

(614)633-5070

Computational engineer specializing in thermal-fluid simulation, multiphysics modeling, and electronics thermal management. Experience developing CFD/FEA models, scalable post-processing workflows, and HPC-enabled simulation tools for semiconductor and thermal transport systems.

TECHNICAL SKILLS

- **Thermal & Fluids:** Electronics cooling, liquid cooling, heat transfer, fluid mechanics, thermodynamics, conjugate heat transfer, multiphysics modeling
- **Simulation & Validation:** CFD, FEA, FEM/FVM, steady-state and transient analysis, meshing, convergence analysis, verification and validation, simulation-to-test correlation
- **Software:** Ansys Fluent, Icepak, Mechanical/APDL, Workbench, Flotherm, VTK
- **Programming and HPC:** Python, C++, Fortran, Bash, multithreading, HPC workflows, performance optimization, automated post-processing and visualization

EXPERIENCE

Thermal Engineer, Supermicro 2026–

- Perform CFD and thermal analysis to evaluate next-generation GPU platforms, including thermal performance and compatibility with Supermicro CDM and CDU liquid cooling systems.
- Evaluate system requirements and thermal solutions for enterprise customers, providing technical support for eight figures in product sales.
- Lead technical coordination with hardware vendors, including product testing, performance evaluation, qualification, and resolution of system integration issues.

Mechanical Engineering Intern, Ansys Summer 2025

- Developed high-performance C++ tools for large-scale thermal and structural FEA datasets, reducing processing time by >80% while improving memory efficiency for multi-gigabyte simulation workflows.
- Developed scalable analysis and visualization workflows for semiconductor chip- and package-level thermal simulations.
- Built multithreaded VTK-based visualization workflows for V&V of large FEA models.

Graduate Research Assistant, Oden Institute (UT Austin) 2023–2026

- Designed and built a computational budgeting framework from first principles, translating conservation equations into scalable tools for diagnosing heat, mass, and momentum transport.
- Took the framework from mathematical formulation through implementation, validation, and HPC-scale analysis, automating previously manual analysis of high-dimensional simulation data.

Graduate Research Assistant, University of Helsinki 2022–2023

- Utilized HPC and FVM to simulate coupled transport processes and validate continuum models against particle-based simulations.
- Built reduced-order models for process identification and engineering interpretation.
- **Publication:** *Polar Science (2025)*

Undergraduate Research Assistant, Bureau of Economic Geology (UT Austin) 2018–2021

- Modeled multiphase transport in porous media using lattice Boltzmann methods.
- Applied parallel computing techniques to accelerate simulation workflows.
- Analyzed interface dynamics and transport behavior in heterogeneous materials.
- **Publication**: *Geophysical Research Letters (2020)*

Graduate Research, Computational Science and Engineering (CFD) — UT Austin
M.Sc., Atmospheric Sciences (CFD) — University of Helsinki (2023)
B.Sc., Geosciences (Honors) — UT Austin Jackson School (2021)